

24K-0826

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COAL – ASSIGNMENT#03

Q1)

Code:

```
INCLUDE Irvine32.inc
```

```
.data
```

```
Dividend  DWORD 0D4A4h
```

```
Divisor   DWORD 0Ah
```

```
Threshold DWORD 5h
```

```
msgDiv    BYTE "Dividend: ",0
```

```
msgDvs    BYTE "Divisor: ",0
```

```
msgQuo    BYTE "Quotient: ",0
```

```
.code
```

```
main PROC
```

```
mov edx, OFFSET msgDiv
```

```
call WriteString
```

```
mov eax, Dividend
```

```
call WriteHex
```

```
call Crlf
```

```
mov edx, OFFSET msgDvs
```

```
call WriteString
```

```
mov eax, Divisor
```

call WriteHex

call Crlf

mov eax, Dividend

mov ebx, Divisor

call CalcRecursiveDiv

mov edx, OFFSET msgQuo

call WriteString

call WriteHex

call Crlf

exit

main ENDP

CalcRecursiveDiv PROC

cmp eax, Threshold

jle BaseCase

sub eax, ebx

call CalcRecursiveDiv

inc eax

ret

BaseCase:

mov eax, 0

ret

CalcRecursiveDiv ENDP

END main

Screenshot:

The screenshot shows the Microsoft Visual Studio IDE with the assembly file 'prac.asm' open. The code includes Irvine32.inc and defines data for Dividend (0000D4A4h), Divisor (0000000Ah), and Threshold (5h). It also defines message strings for Dividend, Divisor, and Quotient. The main procedure calls CalcRecursiveDiv and prints the results. The debug console shows the output: Dividend: 0000D4A4, Divisor: 0000000A, Quotient: 00001544. The process exited with code 0 (0x0).

```
1 INCLUDE Irvine32.inc
2
3 .data
4 Dividend    DWORD 0D4A4h
5 Divisor     DWORD 0Ah
6 Threshold   DWORD 5h
7 msgDiv      BYTE "Dividend: ",0
8 msgDvs      BYTE "Divisor: ",0
9 msgQuo      BYTE "Quotient: ",0
10
11 .code
12 main PROC
13     mov edx, OFFSET msgDiv
14     call WriteString
15     mov eax, Dividend
16     call WriteHex
17     call Crlf
18
19     mov edx, OFFSET msgDvs
20     call WriteString
21     mov eax, Divisor
22     call WriteHex
23     call Crlf
24
25     mov eax, Dividend
26     mov ebx, Divisor
27
28     call CalcRecursiveDiv
29
30     mov edx, OFFSET msgQuo
31     call WriteString
32     call WriteHex
33     call Crlf
34
35     exit
36 main ENDP
37
38 CalcRecursiveDiv PROC
39     cmp eax, Threshold
40     jle BaseCase
41
42     sub eax, ebx
43     call CalcRecursiveDiv
44
45     inc eax
46     ret
47
48 BaseCase PROC
49     ret
50 BaseCase ENDP
51
```

Dividend: 0000D4A4
Divisor: 0000000A
Quotient: 00001544

C:\Users\Muhammad Abu Bakar\source\repos\finprac\Debug\finprac.exe (process 40184) exited with code 0 (0x0).
Press any key to close this window . . .

This is a close-up of the debug console window from the previous screenshot, showing the same output: Dividend: 0000D4A4, Divisor: 0000000A, Quotient: 00001544, and the process exit message.

```
Dividend: 0000D4A4
Divisor: 0000000A
Quotient: 00001544

C:\Users\Muhammad Abu Bakar\source\repos\finprac\Debug\finprac.exe (process 40184) exited with code 0 (0x0).
Press any key to close this window . . .
```

Q2

Code:

```
INCLUDE Irvine32.inc
```

```
.data
```

```
buffer    BYTE 128 DUP(0)
```

```
byteCount DWORD ?
```

```
cntA      DWORD 0
```

```
cntE      DWORD 0
```

```
cntI      DWORD 0
```

```
cntO      DWORD 0
```

```
cntU      DWORD 0
```

```
prompt    BYTE "Enter a string: ",0
```

```
strHeader BYTE "Vowel Count",0
```

```
outA      BYTE "a or A = ",0
```

```
outE      BYTE "e or E = ",0
```

```
outI      BYTE " i or I = ",0
```

```
outO      BYTE " o or O = ",0
```

```
outU      BYTE " u or U = ",0
```

```
tabSpace  BYTE " ",0
```

```
.code
```

```
main PROC
```

```
mov edx, OFFSET prompt
```

```
call WriteString
```

```
mov edx, OFFSET buffer
```

```
mov ecx, SIZEOF buffer
```

```
call ReadString
mov byteCount, eax
```

```
mov esi, OFFSET buffer
mov ecx, byteCount
```

```
cmp ecx, 0
je PrintResults
```

```
ParseLoop:
mov al, [esi]
```

```
cmp al, 'A'
je FoundA
cmp al, 'a'
je FoundA
```

```
cmp al, 'E'
je FoundE
cmp al, 'e'
je FoundE
```

```
cmp al, 'I'
je FoundI
cmp al, 'i'
je FoundI
```

```
cmp al, 'O'
je FoundO
```

```
cmp al, 'o'  
je FoundO
```

```
cmp al, 'U'  
je FoundU  
cmp al, 'u'  
je FoundU
```

```
jmp NextChar
```

```
FoundA:  
inc cntA  
jmp NextChar
```

```
FoundE:  
inc cntE  
jmp NextChar
```

```
FoundI:  
inc cntI  
jmp NextChar
```

```
FoundO:  
inc cntO  
jmp NextChar
```

```
FoundU:  
inc cntU  
jmp NextChar
```

```
NextChar:  
inc esi  
loop ParseLoop
```

PrintResults:

call Crlf

mov edx, OFFSET strHeader

call WriteString

call Crlf

mov edx, OFFSET outA

call WriteString

mov eax, cntA

call WriteDec

mov edx, OFFSET outI

call WriteString

mov eax, cntI

call WriteDec

mov edx, OFFSET outU

call WriteString

mov eax, cntU

call WriteDec

call Crlf

mov edx, OFFSET outE

call WriteString

mov eax, cntE

call WriteDec

mov edx, OFFSET outO

call WriteString

mov eax, cntO

call WriteDec

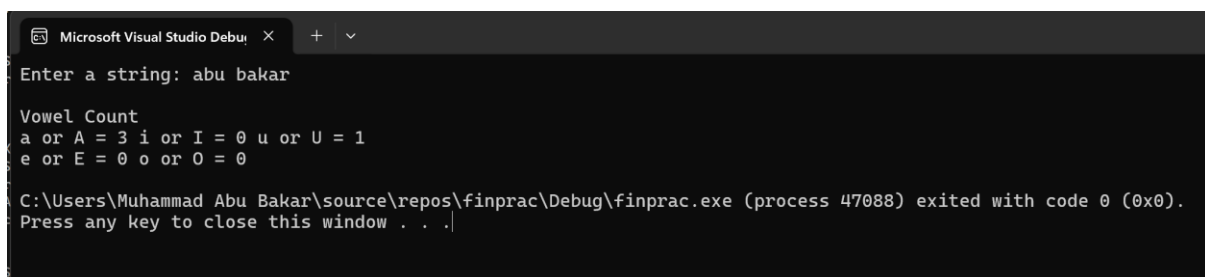
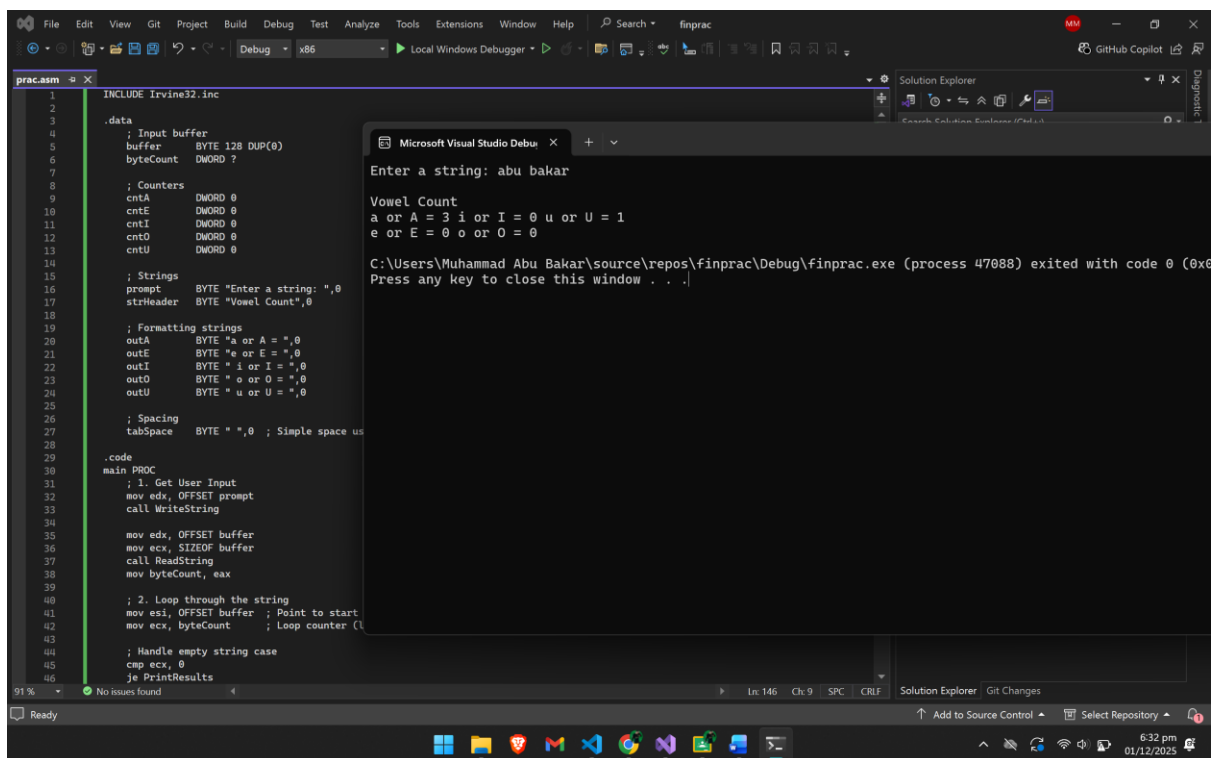
call CrLf

exit

main ENDP

END main

Screenshot:



Q3)**Code:**

```
INCLUDE Irvine32.inc
```

```
.data
```

```
myString BYTE "###FAST",0
```

```
charToTrim BYTE "#"
```

```
msgOriginal BYTE "Original string: ",0
```

```
msgTrimmed BYTE "Trimmed string: ",0
```

```
.code
```

```
main PROC
```

```
mov edx, OFFSET msgOriginal
```

```
call WriteString
```

```
mov edx, OFFSET myString
```

```
call WriteString
```

```
call Crlf
```

```
mov edx, OFFSET myString
```

```
mov al, charToTrim
```

```
call Str_trim_leading
```

```
mov edx, OFFSET msgTrimmed
```

```
call WriteString
```

```
mov edx, OFFSET myString
```

```
call WriteString
```

```
call Crlf
```

exit

main ENDP

Str_trim_leading PROC USES eax ebx esi edi

mov esi, edx

mov edi, edx

mov bl, al

ScanLoop:

mov al, [esi]

cmp al, 0

je StartCopy

cmp al, bl

jne StartCopy

inc esi

jmp ScanLoop

StartCopy:

CopyLoop:

mov al, [esi]

mov [edi], al

cmp al, 0

je Done

inc esi

inc edi

jmp CopyLoop

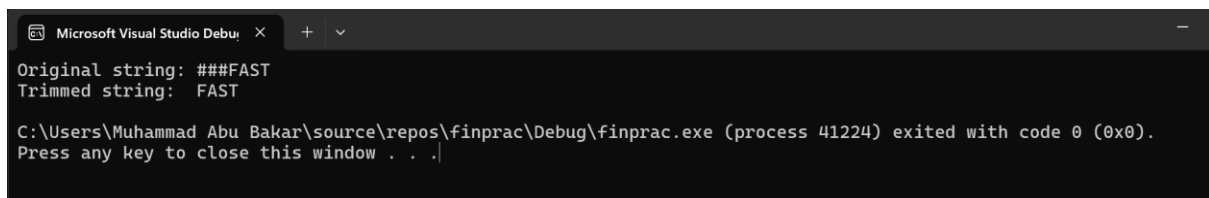
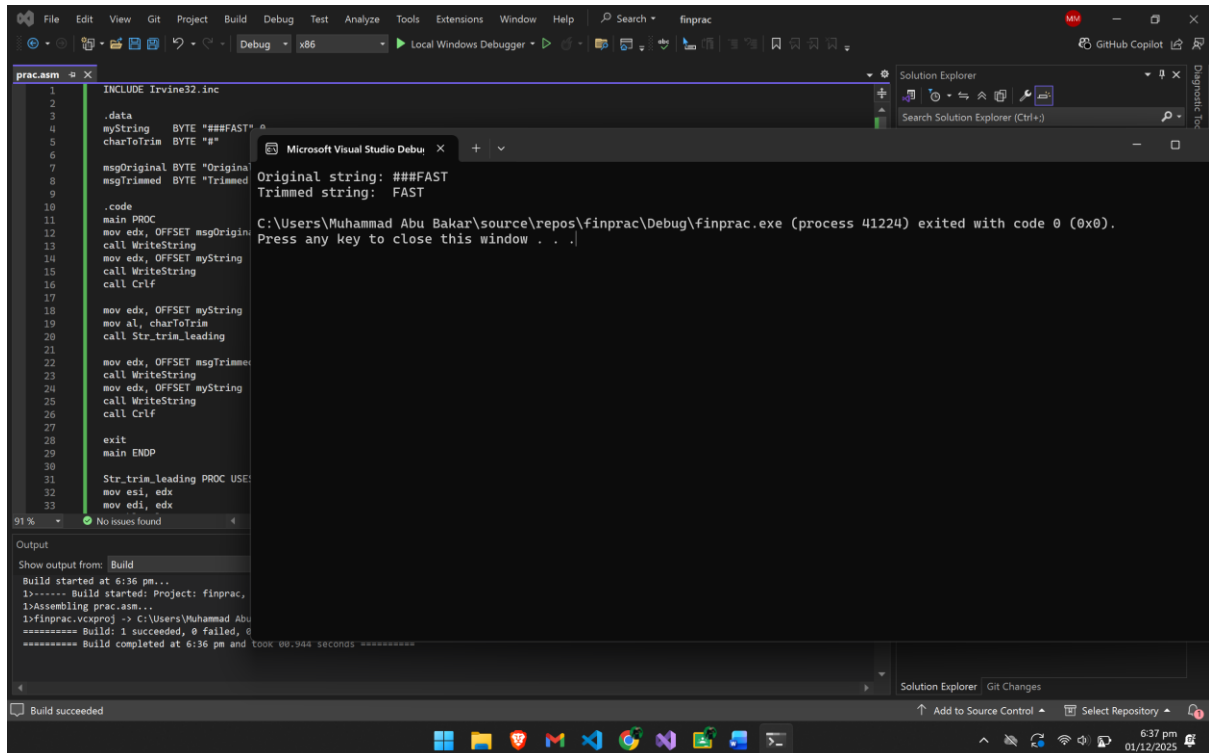
Done:

ret

Str_trim_leading ENDP

END main

Screenshot:



Q4)

Code:

```
INCLUDE Irvine32.inc
```

```
FindFive PROTO :PTR SDWORD, :DWORD
```

```
.data
```

```
array1    SDWORD 5, 5, 5, 5, 5, 10, 20
```

```
array2    SDWORD 1, 2, 5, 5, 5, 5, 5, 3
```

```
array3    SDWORD 5, 5, 5, 0, 5, 5, 5
```

```
array4    SDWORD 1, 2, 3, 4, 6, 7, 8
```

```
strTest1  BYTE "Test 1 (Start): ",0
```

```
strTest2  BYTE "Test 2 (Middle): ",0
```

```
strTest3  BYTE "Test 3 (Broken): ",0
```

```
strTest4  BYTE "Test 4 (None):  ",0
```

```
strPass   BYTE "Found sequence!",0
```

```
strFail   BYTE "No sequence found.",0
```

```
.code
```

```
main PROC
```

```
mov edx, OFFSET strTest1
```

```
call WriteString
```

```
INVOKE FindFive, ADDR array1, LENGTHOF array1
```

```
call ShowResult
```

```
mov edx, OFFSET strTest2
```

```
call WriteString
```

```
INVOKE FindFive, ADDR array2, LENGTHOF array2
```

```
call ShowResult
```

```
mov edx, OFFSET strTest3
```

```
call WriteString
```

```
INVOKE FindFive, ADDR array3, LENGTHOF array3
```

```
call ShowResult
```

```
mov edx, OFFSET strTest4
```

```
call WriteString
```

```
INVOKE FindFive, ADDR array4, LENGTHOF array4
```

```
call ShowResult
```

```
exit
```

```
main ENDP
```

```
ShowResult PROC
```

```
cmp eax, 1
```

```
je IsFound
```

```
mov edx, OFFSET strFail
```

```
jmp DoPrint
```

```
IsFound:
```

```
mov edx, OFFSET strPass
```

```
DoPrint:
```

```
call WriteString
```

```
call Crlf
```

```
call Crlf
```

```
ret
```

```
ShowResult ENDP
```

FindFive PROC USES esi ecx ebx,

ptrArray:PTR SDWORD,

arrSize:DWORD

mov esi, ptrArray

mov ecx, arrSize

mov ebx, 0

cmp ecx, 5

jb NotFound

ScanLoop:

mov eax, [esi]

cmp eax, 5

jne ResetCount

inc ebx

cmp ebx, 5

je Found

jmp NextIter

ResetCount:

mov ebx, 0

NextIter:

add esi, 4

loop ScanLoop

NotFound:

```
mov eax, 0  
  
jmp TheEnd
```

Found:

```
mov eax, 1
```

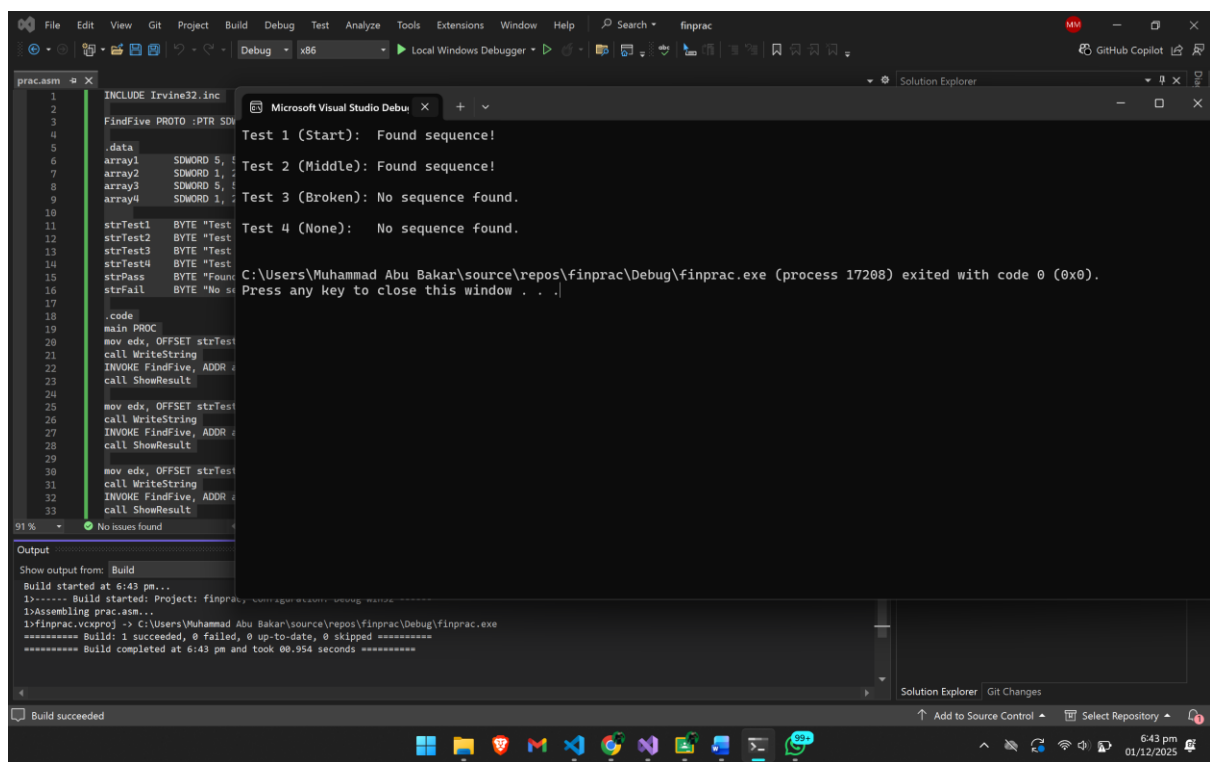
TheEnd:

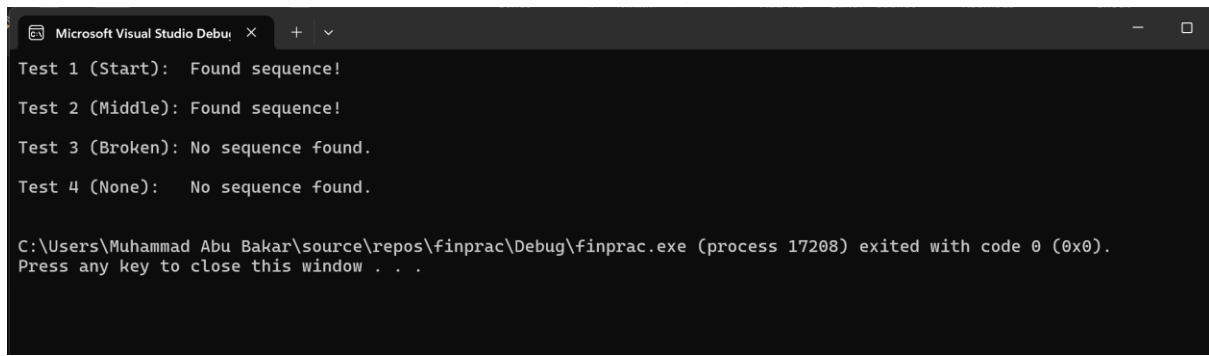
```
ret
```

```
FindFive ENDP
```

```
END main
```

Screenshot:





The image shows a screenshot of a Microsoft Visual Studio Debug Console window. The window has a dark theme and a title bar that reads "Microsoft Visual Studio Debug". Inside the console, there are four lines of output representing test results: "Test 1 (Start): Found sequence!", "Test 2 (Middle): Found sequence!", "Test 3 (Broken): No sequence found.", and "Test 4 (None): No sequence found.". Below these, a message indicates that the process "C:\Users\Muhammad Abu Bakar\source\repos\finprac\Debug\finprac.exe (process 17208)" has exited with code 0 (0x0), followed by "Press any key to close this window . . .".

```
Microsoft Visual Studio Debug
Test 1 (Start): Found sequence!
Test 2 (Middle): Found sequence!
Test 3 (Broken): No sequence found.
Test 4 (None): No sequence found.

C:\Users\Muhammad Abu Bakar\source\repos\finprac\Debug\finprac.exe (process 17208) exited with code 0 (0x0).
Press any key to close this window . . .
```